

Lab3 Reflection

Question #1

Explain why I used local variables

I used local variables in my code for the following reasons:

- It helps prevent accidental overrides or naming conflicts in large programs. This happens because any variable declared without the local scope keyword are global, i.e. accessible from anywhere in the program.
- It also helps cut down on memory use by allowing them to be garbage collected when they go out of scope
- Finally it helps improves performance by removing the need to look up values in the G table.

Question 2

Explain how the __index metamethod functioned in part 2

The __index metamethod was used to lookup a key that did not exist in the SpecialItem table. Since we used the setMetaTable function to set BaseTemplate's metatable as a fall back for SpecialItem, the lua interpreter will also check to see if the key exist in the __index. This allows us to create tables that inherit from other tables.